Average Rate of Change

ANSWER KEY

Section 1: Formula and Secant Line Concept

1 State the formula for the average rate of change of $f(x)$ over $[a, b]$.

a. $\frac{f(b)-f(a)}{b-a}$ ✓

b. $\frac{f(a)+f(b)}{2}$

c. $\frac{f(b)-f(a)}{2}$

d. $\frac{b-a}{f(b)-f(a)}$

The average rate of change is the slope of the secant line passing through $(a, f(a))$ and $(b, f(b))$: $\frac{f(b)-f(a)}{b-a}$.

2 Find the average rate of change of $f(x) = x^2 - 3x$ from $x = 1$ to $x = 4$.

a. 2 ✓

b. 4

c. 3

d. 1

$f(1) = 1 - 3 = -2$. $f(4) = 16 - 12 = 4$. Rate $= \frac{4-(-2)}{4-1} = \frac{6}{3} = 2$.

3 Find the slope of the secant line connecting $(2, f(2))$ and $(5, f(5))$ for $f(x) = 2x^2 + 1$.

a. 14 ✓

b. 12

c. 16

d. 10

$f(2) = 9$, $f(5) = 51$. Slope $= \frac{51-9}{5-2} = \frac{42}{3} = 14$.

Section 2: Tabular and Data-Based Rates of Change

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4 A table shows $t = [0, 2, 5]$ and distance $s(t) = [0, 24, 75]$. What is the average velocity from $t = 2$ to $t = 5$?

a. 17 m/s ✓

b. 15 m/s

c. 25 m/s

d. 12 m/s

Average velocity $= \frac{s(5) - s(2)}{5 - 2} = \frac{75 - 24}{3} = \frac{51}{3} = 17$ m/s.

5 Water depth in a reservoir drops from 80 ft at day 4 to 68 ft at day 10. Find the average rate of change in depth per day.

a. -2 ft/day ✓

b. 2 ft/day

c. -1.2 ft/day

d. -12 ft/day

$\frac{68 - 80}{10 - 4} = \frac{-12}{6} = -2$ ft/day.

6 If the average rate of change of $g(x)$ over $[1, 6]$ is 4 and $g(1) = 5$, find $g(6)$.

a. 25 ✓

b. 20

c. 29

d. 24

$\frac{g(6) - 5}{6 - 1} = 4 \Rightarrow g(6) - 5 = 20 \Rightarrow g(6) = 25$.

Section 3: Algebraic Expressions with Variables

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7 Find the average rate of change of $f(x) = x^2$ over the interval $[2, 2+h]$ in terms of h.

a. $4 + h$ ✓

b. 4

c. $2 + h$

d. $4 + 2h$

$\frac{f(2+h)-f(2)}{h} = \frac{(2+h)^2-4}{h} = \frac{4+4h+h^2-4}{h} = \frac{4h+h^2}{h} = 4+h.$

8 Find the average rate of change of $f(x) = 3x^2 - 5$ over $[x, x+h]$.

a. $6x + 3h$ ✓

b. $6x$

c. $3x + 3h$

d. $6x - 5$

$\frac{3(x+h)^2-5-(3x^2-5)}{h} = \frac{6xh+3h^2}{h} = 6x+3h.$

9 For $f(x) = \frac{1}{x}$, find the average rate of change over $[a, b]$.

a. $-\frac{1}{ab}$ ✓

b. $\frac{1}{ab}$

c. $-\frac{1}{a+b}$

d. $\frac{a-b}{ab}$

$\frac{1/b-1/a}{b-a} = \frac{(a-b)/(ab)}{b-a} = -\frac{b-a}{ab(b-a)} = -\frac{1}{ab}.$

Section 4: Contextual Applications

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10 A projectile's height is given by $h(t) = -16t^2 + 80t$. Find its average velocity over the first 3 seconds $[0, 3]$.

a. 32 ft/s ✓

b. 48 ft/s

c. 16 ft/s

d. 80 ft/s

$h(0) = 0$. $h(3) = -16(9) + 80(3) = -144 + 240 = 96$ ft. Average velocity $= \frac{96 - 0}{3 - 0} = 32$ ft/s.

11 A company's revenue in thousands of dollars is $R(x) = 50x - x^2$, where x is in hundreds of units. What is the average rate of change in revenue as production increases from $x = 10$ to $x = 20$?

a. 20 thousand dollars per unit ✓

b. 30

c. 15

d. 25

$R(10) = 500 - 100 = 400$. $R(20) = 1000 - 400 = 600$. Rate $= \frac{600 - 400}{20 - 10} = \frac{200}{10} = 20$.

12 Why does a linear function $f(x) = mx + b$ have the same average rate of change over EVERY interval?

a. **Because the slope of the secant line between any two points is constantly** m ✓

b. Because $b = 0$

c. Because the function has no intercepts

d. Because $f(x)$ is symmetric

For any a and b, $\frac{(mb + b_0) - (ma + b_0)}{b - a} = \frac{m(b - a)}{b - a} = m$, which is constant.

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